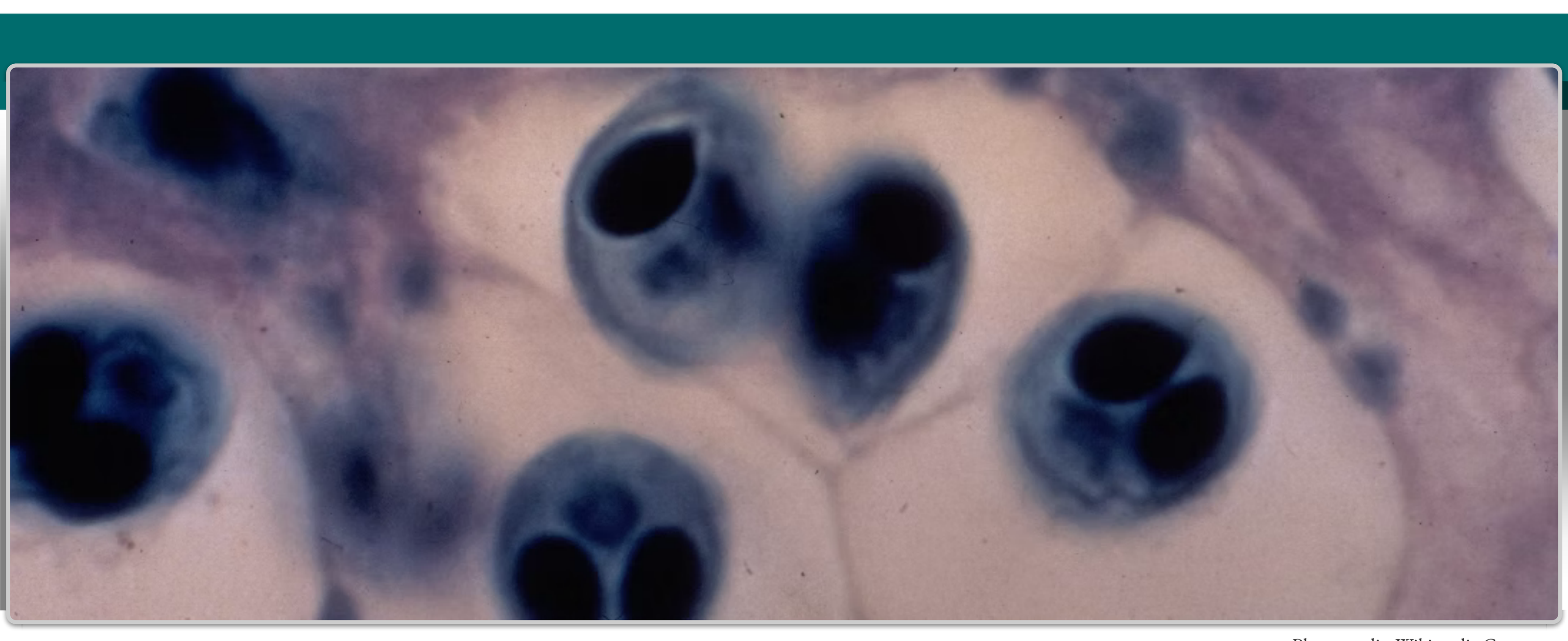

Photo credit: Wikimedia Commons

Whirling Disease

(Myxobolus cerebralis)

Report this Species!

If you believe you have found this species anywhere in Pennsylvania, please report your findings to state and federal authorities immediately!

- **Pennsylvania Fish & Boat Commission** (state authority)
- **U.S. Fish & Wildlife Service** (federal authority)

Species at a Glance

Whirling disease is an infection caused by the microscopic parasite *Myxobolus cerebralis*, which impacts members of the salmonid family, such as trout, salmon, and whitefish. This parasite is an exotic species from Europe that can cause fish to swim in a characteristic "whirl" pattern. It is a significant threat to hatcheries and wild cold-water fisheries in North America.

Identification

This parasitic organism has a complex life cycle requiring two hosts, tubifex worms and the Salmonidae family of fish. Its life begins when tubifex worms consume spores found in the sediment of river bottoms. The worms become infected and the spores develop into a free-swimming form called triactinomyxon (TAM). When the host cell bursts it releases TAM spores into the water. These spores can attach to a fish's skin, penetrate through secretory openings and settle in its head and cartilage. The TAM then multiplies rapidly and eventually invades the spinal cord and the brain and it puts pressure on the organ of equilibrium, causing the fish to swim erratically (whirl). This makes it difficult for the fish to feed and avoid predators. Other physical symptoms include blackened tail and deformities of the head and spine. Infected fish release mature spores into the water, and the cycle begins again.

Similar Species

The symptoms can have characteristics similar to those of other fish diseases, so lab testing is necessary to confirm that a fish is infected.

Habitat

Because spores are released from the tubifex worm almost exclusively when the temperature is 10°-15°C (50-59°F), fish in warmer or cooler waters are less likely to be infected. The spores can withstand freezing, drying, and passing through the digestive tracts of predators. They can also survive in the stream for 20-30 years. Once in the short-lived TAM stage, infection of fish hosts is dependent upon temperature as well as other environmental factors.

Spread

When an infected fish dies, many thousands to millions of the parasite spores are released to the water. The parasite can spread naturally through a watershed, or by humans moving infected fish and fish parts, or moving mud or water from one watershed to another.

Distribution

The parasite, which is native to Europe, was first identified in the Benner Springs fish hatchery in Pennsylvania in the late 1950s, presumably arriving with frozen fish shipments. It has spread to more than 25 states. In Pennsylvania, outbreaks have only occurred in fish hatchery operations, and although the parasite has been detected in wild streams, it has not led to outbreaks in wild trout populations.

Note: Distribution data for this species may have changed since the publication of ***Pennsylvania's Field Guide to Aquatic Invasive Species (Second Edition 2015)****, the source of information for this description.*

Environmental Impacts

All species of trout and salmon may be susceptible to whirling disease, although rainbow and cutthroat trout appear to be the most vulnerable. Brown trout appear to have immunity to the infection and have not been as greatly impacted. Young fish are also more susceptible because their skeletons have not ossified and they are more prone to deformities. Whirling disease cannot infect humans, mammals, or fish that are not members of the salmonid family.

Video

Note

Information for this species profile comes from **Pennsylvania's Field Guide to Aquatic Invasive Species (Second Edition 2015)**.

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